

NIM : 245410034

Nama : Irvan Cahya Nugraha

Kelas : IF-1

LISTING PRAKTIKUM PEMROGRAMAN SISI BACK END

MODUL 7

SQL

PRAKTIK

Praktik 1

```
C:\xampp\mysql\bin>mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 49
Server version: 10.4.32-MariaDB mariadb.org binary distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

```
Administrator: C:\WINDOWS\ × + ∨

MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| datamahasiswa |
| information_schema |
| login |
| mysql |
| online_shop |
| performance_schema |
| phpmyadmin |
| produk |
| test |
| user_db |
+-----+
10 rows in set (0.002 sec)

MariaDB [(none)]> CREATE DATABASE storedb;
Query OK, 1 row affected (0.002 sec)

MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| datamahasiswa |
| information_schema |
| login |
| mysql |
| online_shop |
| performance_schema |
| phpmyadmin |
| produk |
| storedb |
| test |
| user_db |
+-----+
11 rows in set (0.001 sec)

MariaDB [(none)]> DROP DATABASE storedb;
Query OK, 0 rows affected (0.037 sec)

MariaDB [(none)]> CREATE DATABASE IF NOT EXISTS storedb;
Query OK, 1 row affected (0.002 sec)

MariaDB [(none)]> DROP DATABASE IF EXISTS storedb;
Query OK, 0 rows affected (0.002 sec)

MariaDB [(none)]> CREATE DATABASE IF NOT EXISTS storedb;
```

```

MariaDB [(none)]> CREATE DATABASE IF NOT EXISTS storedb;
Query OK, 1 row affected (0.002 sec)

MariaDB [(none)]> SHOW CREATE DATABASE storedb \G
***** 1. row *****
      Database: storedb
Create Database: CREATE DATABASE `storedb` /*!40100 DEFAULT CHARACTER SET utf8mb4 COLLATE utf8mb4_general_ci */
1 row in set (0.001 sec)

MariaDB [(none)]> DROP DATABASE IF EXISTS storedb;
Query OK, 0 rows affected (0.003 sec)

MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| datamahasiswa |
| information_schema |
| login |
| mysql |
| online_shop |
| performance_schema |
| phpmyadmin |
| produk |
| test |
| user_db |
+-----+
10 rows in set (0.001 sec)

MariaDB [(none)]> QUIT;
Bye

C:\xampp\mysql\bin>mysql -u root -p|

```

Praktik 2

```

C:\xampp\mysql\bin>mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 50
Server version: 10.4.32-MariaDB mariadb.org binary distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| datamahasiswa |
| information_schema |
| login |
| mysql |
| online_shop |
| performance_schema |
| phpmyadmin |
| produk |
| test |
| user_db |
+-----+
10 rows in set (0.002 sec)

MariaDB [(none)]> CREATE DATABASE `storedb` ;
Query OK, 1 row affected (0.015 sec)

```

```
MariaDB [mysql]> CREATE USER 'store'@'%' IDENTIFIED BY 'store';
Query OK, 0 rows affected (0.022 sec)

MariaDB [mysql]> GRANT ALL PRIVILEGES ON storedb.* TO 'store'@'%';
Query OK, 0 rows affected (0.002 sec)

MariaDB [mysql]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.016 sec)
```

```
MariaDB [mysql]> SHOW DATABASES;
```

Database
datamahasiswa
information_schema
login
mysql
online_shop
performance_schema
phpmyadmin
produk
storedb
test
user_db

```
11 rows in set (0.002 sec)
```

Praktik 3

```

MariaDB [mysql]> USE storedb;
Database changed
MariaDB [storedb]> SELECT DATABASE();
+-----+
| DATABASE() |
+-----+
| storedb    |
+-----+
1 row in set (0.006 sec)

MariaDB [storedb]> SHOW TABLES;
Empty set (0.001 sec)

MariaDB [storedb]> CREATE TABLE IF NOT EXISTS products (
  -> id INT UNSIGNED NOT NULL AUTO_INCREMENT,
  -> productCode CHAR(3) NOT NULL DEFAULT '',
  -> name VARCHAR(30) NOT NULL DEFAULT '',
  -> quantity INT UNSIGNED NOT NULL DEFAULT 0,
  -> price DECIMAL(7,2) NOT NULL DEFAULT 99999.99,
  -> PRIMARY KEY (id)
  -> );
Query OK, 0 rows affected (0.038 sec)

MariaDB [storedb]> SHOW TABLES;
+-----+
| Tables_in_storedb |
+-----+
| products          |
+-----+
1 row in set (0.001 sec)

MariaDB [storedb]> DESCRIBE products;
+-----+-----+-----+-----+-----+-----+
| Field      | Type                | Null | Key | Default | Extra          |
+-----+-----+-----+-----+-----+-----+
| id         | int(10) unsigned    | NO   | PRI | NULL    | auto_increment |
| productCode | char(3)             | NO   |     |         |                |
| name       | varchar(30)         | NO   |     |         |                |
| quantity   | int(10) unsigned    | NO   |     | 0       |                |
| price      | decimal(7,2)        | NO   |     | 99999.99 |                |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.019 sec)

MariaDB [storedb]> SHOW CREATE TABLE products;

```

Praktik 4

```

MariaDB [storedb]> INSERT INTO products VALUES (1001, 'PEN', 'Pen Red', 5000, 1.23)
;
Query OK, 1 row affected (0.016 sec)

MariaDB [storedb]> INSERT INTO products VALUES (NULL, 'PEN', 'Pen Blue', 8000, 1.25
), (NULL, 'PEN', 'Pen Black', 2000, 1.25);
Query OK, 2 rows affected (0.002 sec)
Records: 2 Duplicates: 0 Warnings: 0

MariaDB [storedb]> INSERT INTO products (productCode, name, quantity, price) VALUES
('PEC', 'Pencil 2B', 10000, 0.48), ('PEC', 'Pencil 2H', 8000, 0.49);
Query OK, 2 rows affected (0.002 sec)
Records: 2 Duplicates: 0 Warnings: 0

MariaDB [storedb]> INSERT INTO products (productCode, name) VALUES ('PEC', 'Pencil
HB');
Query OK, 1 row affected (0.002 sec)

MariaDB [storedb]> SELECT * FROM products;
+-----+-----+-----+-----+-----+
| id    | productCode | name      | quantity | price  |
+-----+-----+-----+-----+-----+
| 1001  | PEN         | Pen Red   | 5000     | 1.23   |
| 1002  | PEN         | Pen Blue  | 8000     | 1.25   |
| 1003  | PEN         | Pen Black | 2000     | 1.25   |
| 1004  | PEC         | Pencil 2B | 10000    | 0.48   |
| 1005  | PEC         | Pencil 2H | 8000     | 0.49   |
| 1006  | PEC         | Pencil HB | 0        | 99999.99 |
+-----+-----+-----+-----+-----+
6 rows in set (0.001 sec)

MariaDB [storedb]> DELETE FROM products WHERE id = 1006;
Query OK, 1 row affected (0.016 sec)

MariaDB [storedb]> |

```

Praktik 5

```
MariaDB [storedb]> SELECT name, price FROM products;
```

name	price
Pen Red	1.23
Pen Blue	1.25
Pen Black	1.25
Pencil 2B	0.48
Pencil 2H	0.49

```
5 rows in set (0.001 sec)
```

```
MariaDB [storedb]> SELECT * FROM products;
```

id	productCode	name	quantity	price
1001	PEN	Pen Red	5000	1.23
1002	PEN	Pen Blue	8000	1.25
1003	PEN	Pen Black	2000	1.25
1004	PEC	Pencil 2B	10000	0.48
1005	PEC	Pencil 2H	8000	0.49

```
5 rows in set (0.000 sec)
```

```
MariaDB [storedb]> SELECT 1+1;
```

1+1
2

```
1 row in set (0.000 sec)
```

```
MariaDB [storedb]> SELECT NOW();
```

NOW()
2026-05-06 10:41:25

```
1 row in set (0.000 sec)
```

```
MariaDB [storedb]> SELECT 1+1, NOW();
```

1+1	NOW()
2	2026-05-06 10:41:25

```
1 row in set (0.000 sec)
```

```
MariaDB [storedb]> SELECT name, price FROM products WHERE price < 1.0;
```

name	price
------	-------

```
Administrator: C:\WINDOWS\ × + ∨

MariaDB [storedb]> SELECT name, price FROM products WHERE price < 1.0;
+-----+-----+
| name      | price |
+-----+-----+
| Pencil 2B | 0.48  |
| Pencil 2H | 0.49  |
+-----+-----+
2 rows in set (0.015 sec)

MariaDB [storedb]> SELECT name, quantity FROM products WHERE quantity <= 2000;
+-----+-----+
| name      | quantity |
+-----+-----+
| Pen Black | 2000     |
+-----+-----+
1 row in set (0.000 sec)

MariaDB [storedb]> SELECT name, price FROM products WHERE productCode = 'PEN';
+-----+-----+
| name      | price |
+-----+-----+
| Pen Red   | 1.23  |
| Pen Blue  | 1.25  |
| Pen Black | 1.25  |
+-----+-----+
3 rows in set (0.000 sec)

MariaDB [storedb]> SELECT name, price FROM products WHERE name LIKE 'PENCIL%';
+-----+-----+
| name      | price |
+-----+-----+
| Pencil 2B | 0.48  |
| Pencil 2H | 0.49  |
+-----+-----+
2 rows in set (0.000 sec)

MariaDB [storedb]> SELECT name, price FROM products WHERE name LIKE 'P__ %';
+-----+-----+
| name      | price |
+-----+-----+
| Pen Red   | 1.23  |
| Pen Blue  | 1.25  |
| Pen Black | 1.25  |
+-----+-----+
3 rows in set (0.000 sec)

MariaDB [storedb]> SELECT * FROM products WHERE quantity >= 5000 AND name LIKE 'Pen %';
+-----+-----+-----+-----+-----+
| id  | productCode | name      | quantity | price |
+-----+-----+-----+-----+-----+
| 1001 | PEN         | Pen Red   | 5000     | 1.23  |
| 1002 | PEN         | Pen Blue  | 8000     | 1.25  |
+-----+-----+-----+-----+-----+
```

```
Administrator: C:\WINDOWS\ ...
2 rows in set (0.000 sec)

MariaDB [storedb]> SELECT * FROM products WHERE quantity >= 5000 AND price < 1.24 AND name LIKE 'Pen %';
+----+-----+-----+-----+-----+
| id | productCode | name   | quantity | price |
+----+-----+-----+-----+
| 1001 | PEN         | Pen Red | 5000     | 1.23  |
+----+-----+-----+-----+
1 row in set (0.001 sec)

MariaDB [storedb]> SELECT * FROM products WHERE NOT (quantity >= 5000 AND name LIKE 'Pen %');
+----+-----+-----+-----+-----+
| id | productCode | name       | quantity | price |
+----+-----+-----+-----+
| 1003 | PEN         | Pen Black | 2000     | 1.25  |
| 1004 | PEC         | Pencil 2B | 10000    | 0.48  |
| 1005 | PEC         | Pencil 2H | 8000     | 0.49  |
+----+-----+-----+-----+
3 rows in set (0.000 sec)

MariaDB [storedb]> SELECT * FROM products WHERE name IN ('Pen Red', 'Pen Black');
+----+-----+-----+-----+-----+
| id | productCode | name       | quantity | price |
+----+-----+-----+-----+
| 1001 | PEN         | Pen Red   | 5000     | 1.23  |
| 1003 | PEN         | Pen Black | 2000     | 1.25  |
+----+-----+-----+-----+
2 rows in set (0.001 sec)

MariaDB [storedb]> SELECT * FROM products WHERE (price BETWEEN 1.0 AND 2.0) AND (quantity BETWEEN 1000 AND 2000);
+----+-----+-----+-----+-----+
| id | productCode | name       | quantity | price |
+----+-----+-----+-----+
| 1003 | PEN         | Pen Black | 2000     | 1.25  |
+----+-----+-----+-----+
1 row in set (0.000 sec)

MariaDB [storedb]> SELECT * FROM products WHERE productCode IS NULL;
Empty set (0.000 sec)

MariaDB [storedb]> SELECT * FROM products WHERE name LIKE 'Pen %' ORDER BY price DESC;
+----+-----+-----+-----+-----+
| id | productCode | name       | quantity | price |
+----+-----+-----+-----+
| 1002 | PEN         | Pen Blue  | 8000     | 1.25  |
| 1003 | PEN         | Pen Black | 2000     | 1.25  |
| 1001 | PEN         | Pen Red   | 5000     | 1.23  |
+----+-----+-----+-----+
3 rows in set (0.000 sec)

MariaDB [storedb]> SELECT * FROM products WHERE name LIKE 'Pen %' ORDER BY price DESC, quantity;
+----+-----+-----+-----+-----+
| id | productCode | name       | quantity | price |
+----+-----+-----+-----+
| 1002 | PEN         | Pen Blue  | 8000     | 1.25  |
| 1003 | PEN         | Pen Black | 2000     | 1.25  |
| 1001 | PEN         | Pen Red   | 5000     | 1.23  |
+----+-----+-----+-----+
3 rows in set (0.000 sec)
```

```
Administrator: C:\WINDOWS\
+ + +
MariaDB [storedb]> SELECT * FROM products WHERE name LIKE 'Pen %' ORDER BY price DESC, quantity;
+----+-----+-----+-----+-----+
| id | productCode | name | quantity | price |
+----+-----+-----+-----+-----+
| 1003 | PEN | Pen Black | 2000 | 1.25 |
| 1002 | PEN | Pen Blue | 8000 | 1.25 |
| 1001 | PEN | Pen Red | 5000 | 1.23 |
+----+-----+-----+-----+-----+
3 rows in set (0.000 sec)

MariaDB [storedb]> SELECT * FROM products ORDER BY RAND();
+----+-----+-----+-----+-----+
| id | productCode | name | quantity | price |
+----+-----+-----+-----+-----+
| 1004 | PEC | Pencil 2B | 10000 | 0.48 |
| 1003 | PEN | Pen Black | 2000 | 1.25 |
| 1001 | PEN | Pen Red | 5000 | 1.23 |
| 1002 | PEN | Pen Blue | 8000 | 1.25 |
| 1005 | PEC | Pencil 2H | 8000 | 0.49 |
+----+-----+-----+-----+-----+
5 rows in set (0.001 sec)

MariaDB [storedb]> SELECT * FROM products ORDER BY price LIMIT 2;
+----+-----+-----+-----+-----+
| id | productCode | name | quantity | price |
+----+-----+-----+-----+-----+
| 1004 | PEC | Pencil 2B | 10000 | 0.48 |
| 1005 | PEC | Pencil 2H | 8000 | 0.49 |
+----+-----+-----+-----+-----+
2 rows in set (0.000 sec)

MariaDB [storedb]> SELECT * FROM products ORDER BY price LIMIT 2, 1;
+----+-----+-----+-----+-----+
| id | productCode | name | quantity | price |
+----+-----+-----+-----+-----+
| 1001 | PEN | Pen Red | 5000 | 1.23 |
+----+-----+-----+-----+-----+
1 row in set (0.000 sec)

MariaDB [storedb]> SELECT id AS ID, productCode AS Code, name AS Description, price AS 'Unit Price' FROM products ORDER BY ID;
+----+-----+-----+-----+
| ID | Code | Description | Unit Price |
+----+-----+-----+-----+
| 1001 | PEN | Pen Red | 1.23 |
| 1002 | PEN | Pen Blue | 1.25 |
| 1003 | PEN | Pen Black | 1.25 |
| 1004 | PEC | Pencil 2B | 0.48 |
| 1005 | PEC | Pencil 2H | 0.49 |
+----+-----+-----+-----+
5 rows in set (0.000 sec)

MariaDB [storedb]> SELECT CONCAT(productCode, ' - ', name) AS 'Product Description', price FROM products;
+-----+-----+
| Product Description | price |
+-----+-----+
| 1001 - Pen Red | 1.23 |
| 1002 - Pen Blue | 1.25 |
| 1003 - Pen Black | 1.25 |
| 1004 - Pencil 2B | 0.48 |
| 1005 - Pencil 2H | 0.49 |
+-----+-----+
5 rows in set (0.000 sec)
```

```
Administrator: C:\WINDOWS\ ...
MariaDB [storedb]> SELECT CONCAT(productCode, ' - ', name) AS 'Product Description', price FROM products;
+-----+-----+
| Product Description | price |
+-----+-----+
| PEN - Pen Red      | 1.23  |
| PEN - Pen Blue     | 1.25  |
| PEN - Pen Black    | 1.25  |
| PEC - Pencil 2B    | 0.48  |
| PEC - Pencil 2H    | 0.49  |
+-----+-----+
5 rows in set (0.001 sec)

MariaDB [storedb]> SELECT price FROM products;
+-----+
| price |
+-----+
| 1.23  |
| 1.25  |
| 1.25  |
| 0.48  |
| 0.49  |
+-----+
5 rows in set (0.000 sec)

MariaDB [storedb]> SELECT DISTINCT price AS 'Distinct Price' FROM products;
+-----+
| Distinct Price |
+-----+
| 1.23           |
| 1.25           |
| 0.48           |
| 0.49           |
+-----+
4 rows in set (0.000 sec)

MariaDB [storedb]> SELECT DISTINCT price, name FROM products;
+-----+-----+
| price | name |
+-----+-----+
| 1.23  | Pen Red |
| 1.25  | Pen Blue |
| 1.25  | Pen Black |
| 0.48  | Pencil 2B |
| 0.49  | Pencil 2H |
+-----+-----+
5 rows in set (0.001 sec)

MariaDB [storedb]> SELECT * FROM products ORDER BY productCode, id;
+-----+-----+-----+-----+-----+
| id   | productCode | name      | quantity | price |
+-----+-----+-----+-----+-----+
| 1004 | PEC         | Pencil 2B | 10000    | 0.48  |
| 1005 | PEC         | Pencil 2H | 8000     | 0.49  |
+-----+-----+-----+-----+-----+
```

```
Administrator: C:\WINDOWS\ ...
MariaDB [storedb]> SELECT * FROM products ORDER BY productCode, id;
+----+-----+-----+-----+-----+
| id | productCode | name      | quantity | price |
+----+-----+-----+-----+-----+
| 1004 | PEC         | Pencil 2B | 10000    | 0.48  |
| 1005 | PEC         | Pencil 2H | 8000     | 0.49  |
| 1001 | PEN         | Pen Red   | 5000     | 1.23  |
| 1002 | PEN         | Pen Blue  | 8000     | 1.25  |
| 1003 | PEN         | Pen Black | 2000     | 1.25  |
+----+-----+-----+-----+-----+
5 rows in set (0.000 sec)

MariaDB [storedb]> SELECT * FROM products GROUP BY productCode;
+----+-----+-----+-----+-----+
| id | productCode | name      | quantity | price |
+----+-----+-----+-----+-----+
| 1004 | PEC         | Pencil 2B | 10000    | 0.48  |
| 1001 | PEN         | Pen Red   | 5000     | 1.23  |
+----+-----+-----+-----+-----+
2 rows in set (0.001 sec)

MariaDB [storedb]> SELECT COUNT(*) AS 'Count' FROM products;
+-----+
| Count |
+-----+
| 5      |
+-----+
1 row in set (0.000 sec)

MariaDB [storedb]> SELECT productCode, COUNT(*) FROM products GROUP BY productCode;
+-----+-----+
| productCode | COUNT(*) |
+-----+-----+
| PEC         | 2        |
| PEN         | 3        |
+-----+-----+
2 rows in set (0.000 sec)

MariaDB [storedb]> SELECT productCode, COUNT(*) AS count FROM products GROUP BY productCode ORDER BY count DESC;
+-----+-----+
| productCode | count |
+-----+-----+
| PEN         | 3      |
| PEC         | 2      |
+-----+-----+
2 rows in set (0.001 sec)

MariaDB [storedb]> SELECT MAX(price), MIN(price), AVG(price), STD(price), SUM(quantity) FROM products;
+-----+-----+-----+-----+-----+
| MAX(price) | MIN(price) | AVG(price) | STD(price) | SUM(quantity) |
+-----+-----+-----+-----+-----+
| 1.25       | 0.48       | 0.940000   | 0.371591   | 33000         |
+-----+-----+-----+-----+-----+
1 row in set (0.001 sec)
```

```
Administrator: C:\WINDOWS\ ...
+-----+-----+
| PEN         | 3        |
| PEC         | 2        |
+-----+-----+
2 rows in set (0.001 sec)

MariaDB [storedb]> SELECT MAX(price), MIN(price), AVG(price), STD(price), SUM(quantity) FROM products;
+-----+-----+-----+-----+-----+
| MAX(price) | MIN(price) | AVG(price) | STD(price) | SUM(quantity) |
+-----+-----+-----+-----+-----+
| 1.25       | 0.48       | 0.940000   | 0.371591   | 33000         |
+-----+-----+-----+-----+-----+
1 row in set (0.001 sec)

MariaDB [storedb]> SELECT productCode, MAX(price) AS 'Highest Price', MIN(price) AS 'Lowest Price' FROM products GROUP BY productCode;
+-----+-----+-----+
| productCode | Highest Price | Lowest Price |
+-----+-----+-----+
| PEC         | 0.49          | 0.48         |
| PEN         | 1.25          | 1.23         |
+-----+-----+-----+
2 rows in set (0.000 sec)

MariaDB [storedb]> SELECT productCode, MAX(price), MIN(price), CAST(AVG(price) AS DECIMAL(7,2)) AS 'Average', CAST(STD(price) AS DECIMAL(7,2)) AS 'Std Dev', SUM(quantity) FROM products GROUP BY productCode;
+-----+-----+-----+-----+-----+-----+
| productCode | MAX(price) | MIN(price) | Average | Std Dev | SUM(quantity) |
+-----+-----+-----+-----+-----+-----+
| PEC         | 0.49       | 0.48       | 0.49    | 0.01     | 18000         |
| PEN         | 1.25       | 1.23       | 1.24    | 0.01     | 15000         |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.001 sec)

MariaDB [storedb]> SELECT productCode AS 'Product Code', COUNT(*) AS 'Count', CAST(AVG(price) AS DECIMAL(7,2)) AS 'Average' FROM products GROUP BY productCode HAVING Count >=3;
+-----+-----+-----+
| Product Code | Count | Average |
+-----+-----+-----+
| PEN         | 3      | 1.24    |
+-----+-----+-----+
1 row in set (0.001 sec)

MariaDB [storedb]> SELECT productCode, MAX(price), MIN(price), CAST(AVG(price) AS DECIMAL(7,2)) AS 'Average', SUM(quantity) FROM products GROUP BY productCode WITH ROLLUP;
+-----+-----+-----+-----+-----+
| productCode | MAX(price) | MIN(price) | Average | SUM(quantity) |
+-----+-----+-----+-----+-----+
| PEC         | 0.49       | 0.48       | 0.49    | 18000         |
| PEN         | 1.25       | 1.23       | 1.24    | 15000         |
| NULL        | 1.25       | 0.48       | 0.94    | 33000         |
+-----+-----+-----+-----+-----+
3 rows in set (0.001 sec)

MariaDB [storedb]>
```

Praktik 5

Praktik 6

```
MariaDB [storedb]> UPDATE products SET price = price * 1.1;
Query OK, 5 rows affected, 5 warnings (0.003 sec)
Rows matched: 5  Changed: 5  Warnings: 5

MariaDB [storedb]> SELECT * FROM products;
+----+-----+-----+-----+-----+
| id | productCode | name      | quantity | price |
+----+-----+-----+-----+-----+
| 1001 | PEN         | Pen Red   | 5000     | 1.35  |
| 1002 | PEN         | Pen Blue  | 8000     | 1.38  |
| 1003 | PEN         | Pen Black | 2000     | 1.38  |
| 1004 | PEC         | Pencil 2B | 10000    | 0.53  |
| 1005 | PEC         | Pencil 2H | 8000     | 0.54  |
+----+-----+-----+-----+-----+
5 rows in set (0.000 sec)

MariaDB [storedb]> UPDATE products SET quantity = quantity - 100 WHERE name = 'Pen Red';
Query OK, 1 row affected (0.002 sec)
Rows matched: 1  Changed: 1  Warnings: 0

MariaDB [storedb]> SELECT * FROM products WHERE name = 'Pen Red';
+----+-----+-----+-----+-----+
| id | productCode | name      | quantity | price |
+----+-----+-----+-----+-----+
| 1001 | PEN         | Pen Red   | 4900     | 1.35  |
+----+-----+-----+-----+-----+
1 row in set (0.000 sec)

MariaDB [storedb]> UPDATE products SET quantity = quantity + 50, price = 1.23 WHERE name = 'Pen Red';
Query OK, 1 row affected (0.002 sec)
Rows matched: 1  Changed: 1  Warnings: 0

MariaDB [storedb]> SELECT * FROM products WHERE name = 'Pen Red';
+----+-----+-----+-----+-----+
| id | productCode | name      | quantity | price |
+----+-----+-----+-----+-----+
| 1001 | PEN         | Pen Red   | 4950     | 1.23  |
+----+-----+-----+-----+-----+
1 row in set (0.001 sec)

MariaDB [storedb]> |
```

Praktik 7

```
MariaDB [storedb]>
MariaDB [storedb]> DELETE FROM products WHERE name LIKE 'Pencil%';
Query OK, 2 rows affected (0.004 sec)
```

```
MariaDB [storedb]> SELECT * FROM products;
+-----+-----+-----+-----+-----+
| id | productCode | name | quantity | price |
+-----+-----+-----+-----+-----+
| 1001 | PEN | Pen Red | 4950 | 1.23 |
| 1002 | PEN | Pen Blue | 8000 | 1.38 |
| 1003 | PEN | Pen Black | 2000 | 1.38 |
+-----+-----+-----+-----+-----+
3 rows in set (0.016 sec)
```

```
MariaDB [storedb]> DELETE FROM products;
Query OK, 3 rows affected (0.002 sec)
```

```
MariaDB [storedb]> SELECT * FROM products;
Empty set (0.001 sec)
```

```
MariaDB [storedb]> |
```

Praktik 8

```
MariaDB [storedb]>
MariaDB [storedb]>
MariaDB [storedb]> LOAD DATA LOCAL INFILE 'd:/myProject/products_in.csv' INTO TABLE products COLUMNS TERMINATED BY ',' LINES TERMINATED BY '\r\n';
Query OK, 0 rows affected, 5 warnings (0.019 sec)
Records: 5 Deleted: 0 Skipped: 5 Warnings: 5

MariaDB [storedb]> SELECT * FROM products;
+-----+-----+-----+-----+-----+
| id | productCode | name | quantity | price |
+-----+-----+-----+-----+-----+
| 1001 | PEN | Pen Red | 5000 | 1.23 |
| 1007 | PEN | Pen Blue | 8000 | 1.25 |
| 1008 | PEN | Pen Black | 2000 | 1.25 |
| 1009 | PEC | Pencil 2B | 10000 | 0.49 |
| 1010 | PEC | Pencil 2H | 8000 | 0.49 |
+-----+-----+-----+-----+-----+
5 rows in set (0.000 sec)

MariaDB [storedb]> SELECT * FROM products INTO OUTFILE 'd:/myProject/products_out.csv' COLUMNS TERMINATED BY ',' LINES TERMINATED BY '\r\n';
ERROR 1086 (HY000): File 'd:/myProject/products_out.csv' already exists
MariaDB [storedb]> SELECT * FROM products INTO OUTFILE 'd:/myProject/products_out.csv' COLUMNS TERMINATED BY ',' LINES TERMINATED BY '\r\n';
Query OK, 5 rows affected, 1 warning (0.002 sec)
```

Praktik 9

```

MariaDB [storedb]> USE storedb;
Database changed
MariaDB [storedb]> DROP TABLE IF EXISTS suppliers;
Query OK, 0 rows affected, 1 warning (0.001 sec)

MariaDB [storedb]> CREATE TABLE suppliers (
  -> supplierID INT UNSIGNED NOT NULL AUTO_INCREMENT,
  -> name VARCHAR(30) NOT NULL DEFAULT '',
  -> phone CHAR(8) NOT NULL DEFAULT '',
  -> PRIMARY KEY (supplierID)
  -> );
Query OK, 0 rows affected (0.041 sec)

MariaDB [storedb]> DESCRIBE suppliers;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| supplierID | int(10) unsigned | NO   | PRI | NULL    | auto_increment |
| name       | varchar(30)      | NO   |     |         |               |
| phone      | char(8)          | NO   |     |         |               |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.021 sec)

MariaDB [storedb]> INSERT INTO suppliers VALUES (501, 'ABC Traders', '88881111'), (502, 'XYZ Company', '88882222'), (503, 'QQ Corp', '88883333');
Query OK, 3 rows affected (0.003 sec)
Records: 3 Duplicates: 0 Warnings: 0

MariaDB [storedb]> SELECT * FROM suppliers;
+-----+-----+-----+
| supplierID | name       | phone |
+-----+-----+-----+
| 501        | ABC Traders | 88881111 |
| 502        | XYZ Company | 88882222 |
| 503        | QQ Corp    | 88883333 |
+-----+-----+-----+
3 rows in set (0.002 sec)

MariaDB [storedb]> ALTER TABLE products ADD COLUMN supplierID INT UNSIGNED NOT NULL;
Query OK, 0 rows affected (0.011 sec)
Records: 0 Duplicates: 0 Warnings: 0

MariaDB [storedb]> DESCRIBE products;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id     | int(10) unsigned | NO   | PRI | NULL    | auto_increment |
| productCode | char(3)         | NO   |     |         |               |
| name    | varchar(30)      | NO   |     |         |               |
| quantity | int(10) unsigned | NO   |     | 0       |               |
| price   | decimal(7,2)     | NO   |     | 99999.99 |               |
| supplierID | int(10) unsigned | NO   |     | NULL    |               |
+-----+-----+-----+-----+-----+-----+

MariaDB [storedb]> UPDATE products SET supplierID = 501;
Query OK, 5 rows affected (0.007 sec)
Rows matched: 5 Changed: 5 Warnings: 0

MariaDB [storedb]> ALTER TABLE products ADD FOREIGN KEY (supplierID) REFERENCES suppliers (supplierID);
Query OK, 5 rows affected (0.007 sec)
Records: 5 Duplicates: 0 Warnings: 0

MariaDB [storedb]> DESCRIBE products;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id     | int(10) unsigned | NO   | PRI | NULL    | auto_increment |
| productCode | char(3)         | NO   |     |         |               |
| name    | varchar(30)      | NO   |     |         |               |
| quantity | int(10) unsigned | NO   |     | 0       |               |
| price   | decimal(7,2)     | NO   |     | 99999.99 |               |
| supplierID | int(10) unsigned | NO   | MUL | NULL    |               |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.008 sec)

MariaDB [storedb]> UPDATE products SET supplierID = 502 WHERE id = 2004;
Query OK, 0 rows affected (0.001 sec)
Rows matched: 0 Changed: 0 Warnings: 0

MariaDB [storedb]> SELECT * FROM products;
+-----+-----+-----+-----+-----+-----+
| id | productCode | name       | quantity | price | supplierID |
+-----+-----+-----+-----+-----+-----+
| 1001 | PEN         | Pen Red   | 5000     | 1.23 | 501        |
| 1007 | PEN         | Pen Blue  | 8000     | 1.25 | 501        |
| 1008 | PEN         | Pen Black | 2000     | 1.25 | 501        |
| 1009 | PEC         | Pencil 2B | 10000    | 0.48 | 501        |
| 1010 | PEC         | Pencil 2H | 8000     | 0.49 | 501        |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.000 sec)

MariaDB [storedb]> SELECT products.name, price, suppliers.name FROM products JOIN suppliers ON products.supplierID = suppliers.supplierID WHERE price < 0.6;
+-----+-----+-----+
| name | price | name |
+-----+-----+-----+
| Pencil 2B | 0.48 | ABC Traders |
| Pencil 2H | 0.49 | ABC Traders |
+-----+-----+-----+
2 rows in set (0.001 sec)

MariaDB [storedb]> SELECT products.name, price, suppliers.name FROM products, suppliers WHERE products.supplierID = suppliers.supplierID AND price < 0.6;
+-----+-----+-----+
| name | price | name |
+-----+-----+-----+
| Pencil 2B | 0.48 | ABC Traders |
| Pencil 2H | 0.49 | ABC Traders |
+-----+-----+-----+

```

LATIHAN

Latihan 1

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TUGAS

Tugas 1

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